Oracle Estate Discovery

Technical Specification & Implementation Guide

For the Oracle-to-Azure Cloud Migration & Modernization Platform

Version 3.0 | April 2026 | Owner: Amit (Stream C)

Includes ETL & Data Integration Domain (Domain 7)

Why AWR Is Not Enough

AWR Captures

- SQL execution stats & plans
- Wait events & I/O metrics
- Tablespace & segment sizes
- CPU & memory utilization
- SGA/PGA advisory
- Buffer cache hit ratios

AWR Misses (Critical Gaps)

- Application tier inventory
- Oracle Forms .fmb/.fmx files
- PL/SQL cross-dependencies
- Middleware topology (WebLogic, SOA)
- Licensing model (ULA/NUP/Processor)
- Integration endpoints (APIs, ETL, ESB)
- Hardware specs, VM clusters, Exadata

Without full estate discovery, TCO calculations are 30-50% inaccurate

Seven Discovery Domains

Infrastructure

Servers, storage, VM clusters, Exadata, RAC, network

Database

Schemas, PL/SQL, performance baselines, dependencies

Applications

EBS, APEX, custom Java, BI, batch jobs, ETL

Forms & Reports

.fmb parsing, complexity scoring, UI mapping

Middleware

WebLogic, SOA Suite, OSB, OHS, ODI, GoldenGate

Licensing

Edition, options, metrics, compliance, optimization

ETL & Data Integration

Scheduled jobs, replication, data flows, transformation

Discovery Architecture

L1: Collection	Discovery Agents | Agentless Scanners | File Upload	Python + PowerShell + JDBC + SSH
L2: Parsing	AWR Parser | FMB Parser | DDL Parser	Python + Node.js + ANTLR4
L3: Analysis	Dependency Mapper | AI Complexity Scorer	.NET Core + Neo4j + Azure OpenAI
L4: Storage	Metadata Repository	Azure SQL + Cosmos DB
L5: API	Discovery REST API	.NET Core 8
L6: Dashboard	Estate Explorer | Dependency Viewer	React + TypeScript

Data flows bottom-up: Collection → Parsing → Analysis → Storage → API → Dashboard

Database Discovery: Schema & PL/SQL

Object Type	Data Points	Azure Target
Tables	Columns, types, constraints, partitions, sizes	Azure SQL / PostgreSQL
Indexes	Type, columns, status, statistics	Target-optimized indexes
Views / MVs	Definition SQL, dependencies, refresh config	View rewrite for target dialect
PL/SQL Packages	Line count, procedures, dependencies	.NET API methods
Stored Procedures	Parameters, DML, exception handling	.NET endpoints / Azure Functions
Triggers	Type, timing, PL/SQL body	App-tier logic or DB triggers
DB Links	Target host, objects accessed	Azure Private Link / VNet
Jobs / Schedulers	Intervals, chains, PL/SQL actions	Azure Functions + Logic Apps
Queues (AQ)	Subscribers, payload types, rates	Azure Service Bus

Infrastructure Discovery Methods

Category	Discovery Method	Tools/Scripts	Data Points Collected
Server Hardware	WMI / SSH / LSHW	Windows: WMI | Linux: LSHW	CPU cores, sockets, threads, RAM, NIC count
Storage	OS commands + APIs	iostat, df, storage APIs	IOPS, throughput, latency, capacity, RAID
Virtualization	vCenter API / OVM CLI	VMware / OVM tools	vCPU, vRAM, overcommit ratios, cluster size
Oracle Exadata	dcli + cellcli	Cell CLI, v$cellevent	Cell config, storage servers, InfiniBand, Flash
RAC Cluster	crsctl + asmcmd	Oracle Grid Infrastructure	Node count, interconnect, voting disks, ASM DGs
Network	Device APIs + config	Switch configs, routing tables	VLANs, firewall rules, load balancers, MTU
Backup	RMAN + OS scripts	RMAN catalog, backup logs	Schedules, retention, tape vs disk, block size
OS	SSH + OS commands	uname, dmesg, systemctl	Distro, kernel, patches, systemd, env vars

Infrastructure Discovery: Build Specification

Agent Component

- Python collector: paramiko, pywinrm, pyvmomi
- Input: host credentials
- Output: JSON inventory

Key SQL Scripts

- v$osstat (CPU/cores/memory)
- v$asm_diskgroup (ASM storage)
- v$parameter (instance config)
- dba_data_files (tablespace)

Azure Mapping Engine

- Maps: cores → VM SKU
- IOPS → disk tier
- RAC → availability zones
- Exadata → dedicated host

Output Schema

- { servers: [{cpu, ram, storage, os}]
- clusters: [{nodes, interconnect}]
- network: [{vlans, rules}] }

Domain 2: Database Discovery Methods

Object Category	Discovery Method	SQL/Tool	Output
Schema Objects	DBA_ catalog views	DBA_OBJECTS, DBA_TABLES	Object inventory by type
PL/SQL Code	DBA_SOURCE + deps	DBA_DEPENDENCIES	Code extraction + complexity
Performance	AWR / ASH / ADDM	DBA_HIST_% views	Execution stats, wait events, I/O
Data Volumes	DBA_SEGMENTS + TABLES	Segment advisor	Sizes, row counts, growth trends
Security	DBA_USERS / ROLES	DBA_SYS_PRIVS	User/role/privilege map
HA/DR	V$DATABASE + V$ARCHIVE	Data Guard views	DataGuard/RAC config
Jobs	DBA_SCHEDULER_JOBS	DBA_SCHEDULER_CHAINS	Scheduled tasks, dependencies
Queues	DBA_QUEUES + AQ views	DBA_QUEUE_SUBSCRIBERS	AQ config, subscribers

Domain 2: Database Build Specification

JDBC Collector

- Java/Python JDBC connector
- Executes 50+ discovery queries
- Handles Oracle 10g through 21c
- Connection: TNS or Easy Connect

PL/SQL Analyzer

- Parses DBA_SOURCE code
- ANTLR4 Oracle grammar
- Detects: DBMS_*, dynamic SQL
- Outputs dependency graph

AWR/ASH Parser

- Reads AWR HTML/text reports
- Extracts: Top SQL, wait events
- I/O stats, SGA advisory
- Fallback: Statspack if unlicensed

Complexity Scorer

- Azure OpenAI scoring (1-10)
- Factors: LOC, Oracle constructs
- Dynamic SQL, dependencies
- Data types, transactions

Domain 3: Application Discovery Methods

Application Type	Discovery Method	Tools	Data Points
Oracle EBS	EBS system tables + AD utils	AD utilities, SQL*Plus	Modules, patches, customizations
Oracle APEX	Workspace views + REST	APEX REST export	Apps, pages, LOVs, processes
Custom Java	JMX + classpath scan	JDBC, classpath scan	WAR/EAR, JDBC pools, ORM
Oracle BI/OBIEE	RPD + web catalog	OBIEE tools, REST API	Subject areas, reports, dashboards
Batch/ETL	DBMS_SCHEDULER + crontab	ODI repository	Job chains, dependencies, volumes
Oracle Reports	.rdf/.rep file scan	File parsing tools	Layouts, queries, parameters
Custom PL/SQL	Schema analysis + tracing	v$session tracing	Entry points, call graphs

Domain 3: Application Build Specification

Session Analyzer

- Queries v$session (module, action)
- Maps SQL execution to app
- Correlates JDBC to app tier
- Generates activity matrix

EBS Discovery

- Queries AD_APPLIED_PATCHES
- FND_APPLICATION tables
- Custom forms/reports/workflows
- Concurrent program catalog

APEX Extractor

- REST API export all workspaces
- Page inventory by type
- Process/computation extraction
- Theme/template catalog

App-to-DB Mapper

- Builds app → schema → objects
- Identifies impact radius
- Migration dependency order
- Data lineage tracking

Domain 4: Forms & Reports Discovery Methods

Artifact Type	Discovery Method	Tools	Outputs
Oracle Forms (.fmb)	Binary FMB parser + XML	FMB Parser, frmf2xml	Objects, PL/SQL, UI tree, complexity
Oracle Forms Runtime	Forms servlet discovery	HTTP discovery, form paths	Running forms, versions, modules
Oracle Reports (.rdf)	RDF file parsing	Report Builder utilities	Query structure, parameters, layout
Report Server	Reports servlet enumeration	HTTP discovery, API scan	Report catalog, execution logs
SRW Programs	Dynamic report discovery	SQL queries on form server	Custom SRW programs, data sources
Triggers & PL/SQL	Form trigger extraction	FMB parser + text analysis	Trigger code, complexity score
Shared Libraries	.pll file scanning	Recursive library parsing	Library inventory, shared objects

Domain 4: Forms Build Specification

FMB Binary Parser

- Node.js FMB parser library
- Extracts form structure
- PL/SQL trigger code
- Returns JSON hierarchy

Form Complexity Scorer

- 1-10 complexity scale
- Factors: blocks, items, triggers
- PL/SQL LOC, dynamic SQL
- Reusability scoring

Form-to-App Mapper

- Correlates .fmb to modules
- Maps form data to tables
- Identifies app dependencies
- Migration feasibility score

Report Converter

- RDF → JSON structure
- Query rewriting for target DB
- Layout metadata extraction
- Scheduling integration points

Domain 5: Middleware & Integration Discovery Methods

Component	Discovery Method	Tools	Data Points
WebLogic Server	WLST scripts + JMX	WLST, Java, JMX API	Domains, clusters, app deployments
SOA Suite / BPEL	SOA console API + MDS	Oracle Enterprise Manager	Composites, services, mediation rules
Oracle Service Bus	OSB console export	WLST + OSB SBCONSOLE	Proxies, business services, routes
Oracle HTTP Server	Config file parsing	mod_oc4j, mod_jk configs	Virtual hosts, load balancer rules
Data Integrator (ODI)	ODI repository queries	JDBC to ODI metadata	Packages, projects, scenarios, scheduling
GoldenGate	GGSCI commands + DBA views	DBA_GOLDENGATE*, ggsci	Trail config, replication flows, lag

Domain 5: Middleware Build Specification

WLST Discovery Agent

- Python wrapper around WLST
- Queries all WL domains
- Application topology extraction
- Cluster + server mapping

SOA Connector

- Enterprise Manager REST API
- Queries MDS repository
- Composite versioning
- Endpoint routing analysis

GoldenGate Mapper

- GGSCI + DBA_GOLDENGATE views
- Replication topology graph
- Trail lag & throughput
- Source/target table mapping

Integration Catalog

- DB links, REST endpoints
- File-based integrations (FTP)
- Message queue subscribers
- Data flow dependency map

Domain 6: Licensing & Compliance Discovery Methods

Category	Discovery Method	Data Sources	Outputs
Oracle Edition	v$version + DBA_REGISTRY	Oracle views, license file	EE/SE/Personal/Express detection
Options & Packs	DBA_FEATURE_USAGE_STATISTICS	License.xml, usage views	List of licensed options, usage %
Metric Type	License metadata + config	Oracle licensing files	Processor/NUP/ULA detection
Feature Usage	Feature usage stats table	DBA_FEATURE_USAGE_STATISTICS	Used/unused features, downgrade risk
Virtualization	vCenter + Oracle vmfacts	VMware + OS detection	VM soft partitioning, licensing impact
Support Level	Support contracts query	Oracle CRM, UXIC data	Premier/Extended/Sustaining status

Domain 6: Licensing Build Specification

License Detector

- Queries v$version, DBA_REGISTRY
- Edition + option detection
- License.xml parser
- Returns: edition, options, packs

Feature Usage Scanner

- DBA_FEATURE_USAGE_STATISTICS
- 90-day rolling window analysis
- Usage % by feature
- Downgrade opportunity flagging

Virtualization Impact

- vCenter API integration
- Oracle vmfacts tool
- Soft partitioning detection
- Licensing impact calculation

Compliance Report

- TCO baseline vs. negotiated
- Optimization opportunities
- ULA vs. BYOL recommendation
- Azure licensing pathway

Domain 7: ETL & Data Integration - Discovery Methods

ETL Category	Discovery Method	Key Queries/Tools	Priority
Oracle Scheduled Jobs	DBA_SCHEDULER_JOBS query	job_name, job_action, repeat_interval, state, run_count	Critical
Legacy DBMS_JOB	DBA_JOBS query	job, schema_user, interval, what, broken status	Critical
Oracle Replication	GoldenGate/Streams scan	DBA_GOLDENGATE_*, DBA_MVIEWS, CDC config	Critical
Third-Party ETL Tools	Config file + service scan	.dtsx, .kjb/.ktr, Informatica repo, process scan	High
Custom Scripts	Crontab + file system scan	crontab -l, /etc/cron.d/*, shell/Python script grep	High
Database Links	DBA_DB_LINKS query	owner, db_link, host, active connections	High
External Tables	DBA_EXTERNAL_TABLES query	table_name, directory, access_parameters	Medium
Data Feeds	Network + file watch analysis	netstat port scan, FTP directories, MQ/Kafka topics	Medium

Domain 7: ETL & Data Integration - Build Specification

ETL Job Scanner

- Python + cx_Oracle
- Sprint 3
- Scheduled jobs, DBMS_JOB, DBMS_SCHEDULER

Crontab Harvester

- Python + paramiko
- Sprint 3
- SSH-based cron scan for all OS users

Windows Task Scanner

- Python + pywinrm
- Sprint 3
- WMI scheduled tasks & ETL services

Config File Parser

- Node.js + fast-xml-parser
- Sprint 4
- SSIS .dtsx, Talend .kjb, ODI exports

Connection String Extractor

Data Lineage Builder

Next Steps

1. **Review & Finalize Spec** — All
   Team review of this specification, identify gaps, get sign-off

2. **Build Discovery Agent Scaffold** — Amit
   Python agent framework with JDBC connector + file upload endpoint

3. **Integrate FMB Parser** — Amit + Dharam
   Connect the existing Node.js FMB parser into the Forms Discovery pipeline

4. **Define Metadata Repository Schema** — Amit + Bhupesh
   Azure SQL schema for all 7 discovery domains, feed TCO calculator

5. **Customer Pilot** — Full team
   Run discovery on first customer Oracle estate, validate accuracy

GA Target: July 2026 | 3 Months | 6 Sprints | +2 Slides: Domain 7 (ETL)